# Module 1 — Data Collection & Processing

Data Visualisation in R — Five Questions & Graphs

*E-Commerce Customer Behavior Analysis Using R* ***Kayyala Vinay (192524278)***

## Q1. Which payment method is used most frequently across all orders?

**Graph type:** Bar chart **R function used:** `barplot()`

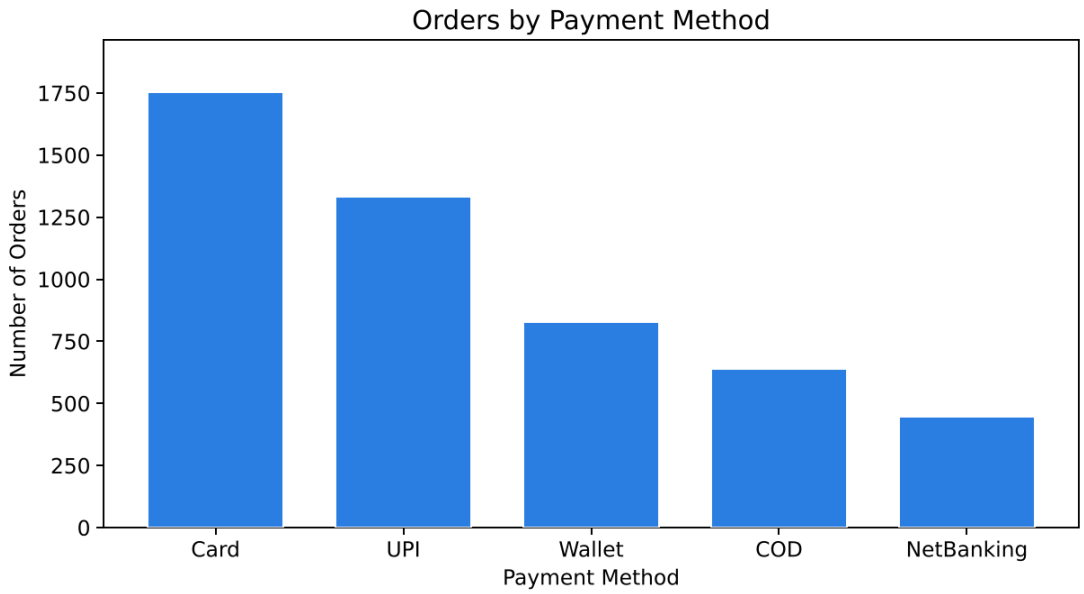
**R code used:**

```
df <- read.csv("C:/Users/kj583/Downloads/archive/cleaned_data.csv")

# Count orders per payment method, highest first
pay <- table(df$payment_method)
pay <- sort(pay, decreasing = TRUE)

# Bar chart
barplot(pay,
        main = "Orders by Payment Method",
        xlab = "Payment Method", ylab = "Number of Orders",
        col  = "#2A7DE1", border = "white")
```

**Output graph:**




## Q2. How does total revenue trend across the months of the year?

**Graph type:** Line chart **R function used:** `plot(..., type = "o")` + `aggregate()`

**R code used:**

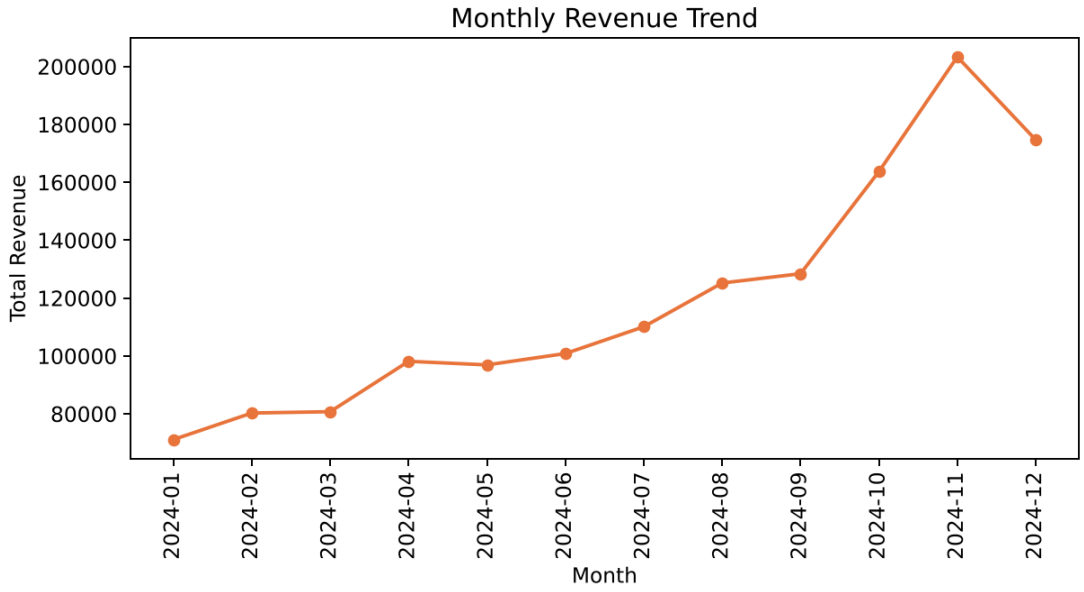
```
df <- read.csv("C:/Users/kj583/Downloads/archive/cleaned_data.csv")
df$order_date <- as.Date(df$order_date)
df$month <- format(df$order_date, "%Y-%m")

# Total revenue per month
monthly <- aggregate(revenue ~ month, data = df, FUN = sum)

# Line chart
plot(monthly$revenue, type = "o", pch = 16, lwd = 2,
     col = "#E8743B", xaxt = "n",
```

```
     main = "Monthly Revenue Trend",
     xlab = "Month", ylab = "Total Revenue")
axis(1, at = 1:nrow(monthly), labels = monthly$month, las = 2)
```

**Output graph:**




## Q3. What share of orders comes from each product category?

**Graph type:** Pie chart **R function used:** `pie()`

**R code used:**

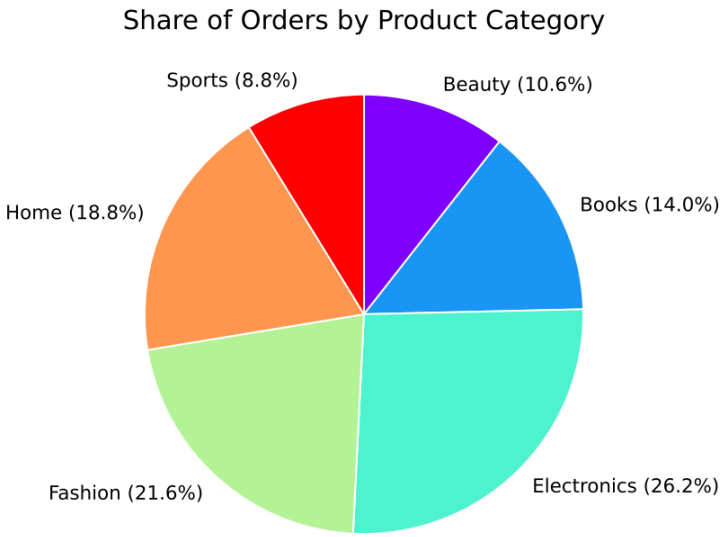
```
df <- read.csv("C:/Users/kj583/Downloads/archive/cleaned_data.csv")

cat_counts <- table(df$product_category)

# Percentage labels
pct <- round(100 * cat_counts / sum(cat_counts), 1)
labels <- paste0(names(cat_counts), " (", pct, "%)")

# Pie chart
pie(cat_counts, labels = labels,
    main = "Share of Orders by Product Category",
    col = rainbow(length(cat_counts)))
```

**Output graph:**




## Q4. How is the unit price distributed after data cleaning?

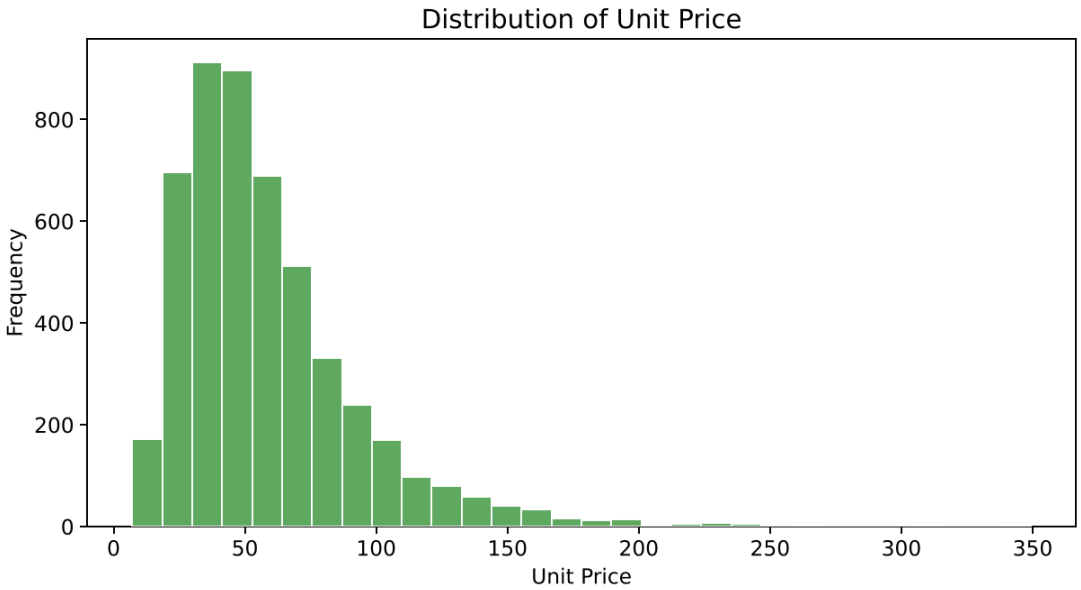
**Graph type:** Histogram **R function used:** `hist()`

**R code used:**

```
df <- read.csv("C:/Users/kj583/Downloads/archive/cleaned_data.csv")

# Histogram of unit_price (30 bins)
hist(df$unit_price,
     breaks = 30,
     main = "Distribution of Unit Price",
     xlab = "Unit Price", ylab = "Frequency",
     col = "#5FA85F", border = "white")
```

**Output graph:**



## Q5. What is the relationship between quantity ordered and revenue?

**Graph type:** Scatter plot (with linear trend line) **R function used:** `plot()` + `abline(lm())`

**R code used:**

```
df <- read.csv("C:/Users/kj583/Downloads/archive/cleaned_data.csv")

# Scatter plot of quantity vs revenue
plot(df$quantity, df$revenue,
     pch = 19, col = rgb(0.16, 0.49, 0.72, 0.4),
     main = "Quantity vs Revenue",
     xlab = "Quantity", ylab = "Revenue")

# Add a linear trend line (least-squares fit)
abline(lm(revenue ~ quantity, data = df), col = "red", lwd = 2)
```

**Output graph:**

